

5 (a) State Kirchhoff's first law.

.....
..... [1]

(b) Fig. 5.1 shows a circuit containing a thermistor T that has a negative temperature coefficient.

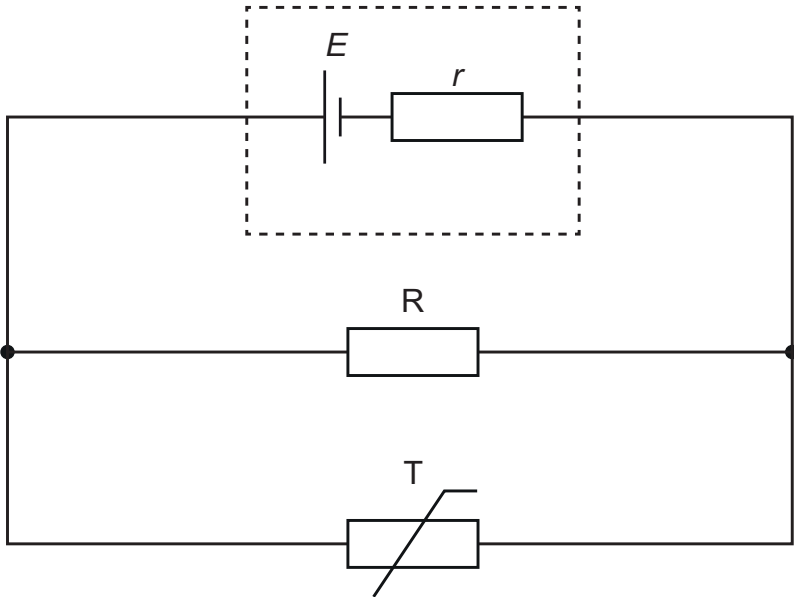


Fig. 5.1

(i) The thermistor has resistance R_0 at a temperature of 0°C .

On Fig. 5.2, sketch a possible variation of the resistance of the thermistor with temperature between 0°C and 100°C .

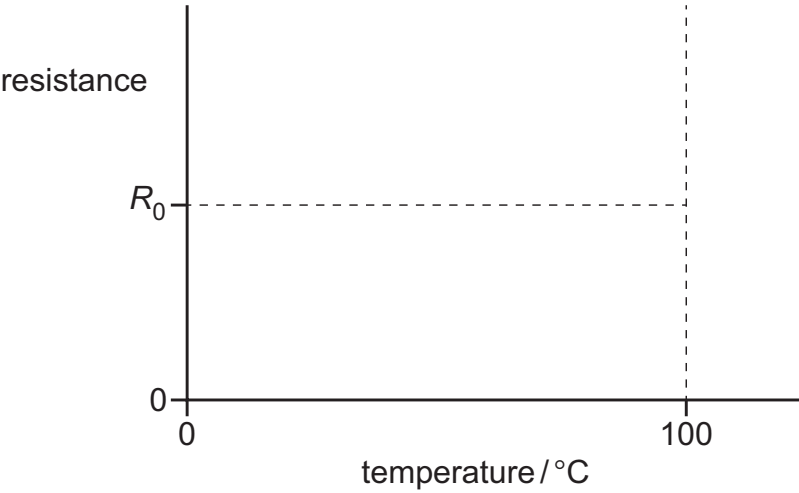


Fig. 5.2

[2]

(ii) With reference to the current in the cell, explain why the current in **resistor R** decreases with increasing temperature of the thermistor.

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..... [3]

(c) The electromotive force (e.m.f.) E of the cell in Fig. 5.1 is 1.50V . The internal resistance r of the cell is $0.12\,\Omega$.

Resistor R has a resistance of $6.00\,\Omega$.

At a particular temperature of the thermistor, the current in R is 0.200A .

For this temperature of the thermistor, determine:

(i) the current in the cell

current =A [2]

(ii) the resistance of the thermistor.

resistance = Ω [2]